Sri Lanka Institute of Information Technology
B. Sc Degree in IT/IS/CSN, Diploma in Information Technology
Year 01 – Semester I – 2017
Mathematics for Computing (IT1030)
Tutorial 11

1. Solve the following systems of linear equations using Cramer's rule:

a) $$\begin{aligned} x_1 + x_3 &= 1 \\ x_2 - x_3 &= 3 \\ 2x_1 + x_2 &= -1 \end{aligned}$$

b) $$\begin{aligned} x_1 + 7x_2 + x_3 &= 1 \\ x_2 - x_3 &= 3 \\ 2x_1 + x_2 + 10x_3 &= -1 \end{aligned}$$

c) $$\begin{aligned} x_1 + x_2 + x_3 &= 1, \\ ax_1 + bx_2 + cx_3 &= d, \\ a^2x_1 + b^2x_2 + c^2x_3 &= d^2, \end{aligned}$$

2. Show that the following sets of equations are inconsistent.

a) $x_1 + 2x_2 + x_3 = 3,$
$x_1 - 3x_2 + 2x_3 = 4,$
$5x_1 + 5x_2 + 6x_3 = 1;$

b) $x_1 + x_2 + x_3 = 2,$
$x_1 + x_3 + 2x_4 = 3,$
$x_1 + x_2 + x_4 = 4,$
$-x_2 + 2x_3 = 2,$

c) $x_1 + x_2 + x_3 - x_4 = 10,$
$x_1 - x_2 - x_3 = 1,$
$4x_1 - 2x_2 - 2x_3 - x_4 = 5.$

3. Find the value of a for which the linear equations

$ax - y + 2z = 1,$
$x + 2y - az = 2,$
$4x + y - 2z = 2,$

has no solutions.

4. Determine the complete set of values for a and b that make the equations

$x + y - z = 2$
$2x + 3y + z = 3$
$5x + 7y + az = b$

i) has a unique solution ii) no solutions iii) an infinite set of solutions.

5. Find the inverses of the following matrices.

a) $\begin{bmatrix} 6 & -3 & 6 \\ 3 & 6 & 6 \\ -12 & -3 & 6 \end{bmatrix}$ b) $\begin{bmatrix} 1 & -1 & 2 \\ 1 & 2 & 1 \\ -4 & -1 & 2 \end{bmatrix}$

7. Find all solution of determinant equation

$$\begin{vmatrix} 1-k & 2 & -1 \\ 2 & 1-k & -1 \\ -1 & -1 & 2-k \end{vmatrix} = 0.$$

What are the values of k for which the following set of equations has nontrivial solutions?

$$\begin{aligned} (1-k)x + \quad & 2y \quad & -z = 0 \\ 2x + \quad & (1-k)y \quad & -z = 0 \\ -x - \quad & y \quad & +(2-k)z = 0 \end{aligned}$$

6. Find the solutions of the following linear equations using the results from question 5.

a)
$$\begin{aligned} 6x - 3y + 6z &= 1 \\ 3x + 6y + 6z &= 5 \\ -12x - 3y + 6z &= 2 \end{aligned}$$

b)
$$\begin{aligned} 4 - y + 2z &= 3 \\ x + 2y + z &= 3 \\ -4x - y + 2z &= 1 \end{aligned}$$